SOFTWARE CHANGE IMPLEMENTATION REPORT

JFICT / CR-009: Reporting and Analytics Endpoints

Field	Details
Change Request ID	CR-009
Change Request Title	Reporting and Analytics Endpoints
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	Completed
Date	22 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators need reliable reports about courses, students, and enrollments, while students need a personal summary of learning activity. Server-side aggregation provides consistent values and avoids calculating totals from incomplete or paginated client data.

1.2 Existing Behaviour

- Before this change, the administrator dashboard did not provide a consolidated reporting endpoint and derived reporting values from separately loaded records in the browser.
- The student dashboard did not receive a dedicated, server-calculated learning report for the authenticated student.
- No role-specific reporting endpoints or consolidated category and top-course breakdowns were available.

1.3 Requested Change

- Provide an administrator-only overview containing system totals, enrollment categories, top courses, courses without enrollments, and recent enrollments.
- Provide a student-only report containing the authenticated student?s enrollment count, category count, earliest/latest dates, total course value, and recent enrollments.
- Calculate report values using SQL aggregation and grouped queries, without issuing one query per course.
- Update both dashboards, handle empty results and reporting failures, and use CSS/table visualizations without adding a chart library.

2. Implementation

2.1 Implementation Summary

Implemented two authenticated reporting endpoints. The admin overview uses administrator role middleware. The student report uses student role middleware and scopes both report queries to the verified JWT user ID (req.user.id), so the client cannot request another student?s report. SQL aggregates and grouped queries calculate report data. Both dashboards consume their role-specific endpoints and provide empty/error handling.

2.2 Affected Files

File	Change	Purpose
backend/server.js	Modified	Mounts report routes under /api/reports.
backend/routes/reportRoutes.js	Added	Defines administrator-only overview and student-only self-report routes.
backend/controllers/reportController.js	Added	Formats report values and returns explicit report errors.
backend/models/reportModel.js	Added	Runs aggregate, category, top-five, and recent-enrollment SQL queries.
frontend/src/pages/AdminDashboard.jsx	Modified	Displays admin report cards, category bars, top courses, and recent enrollments.
frontend/src/pages/StudentDashboard.jsx	Modified	Displays student learning metrics and recent enrollments.
frontend/src/App.css	Modified	Styles category bars and learning activity.

2.3 Technical Changes

Admin Report: GET /api/reports/overview

- Returns total courses, students, and enrollments; students with enrollments; enrollments during the last 30 days; and courses without enrollments.
- Returns category enrollment counts and up to five courses ordered by enrollment count descending.
- Returns the six most recent enrollments for the administrator dashboard.
- Uses aggregate SQL, LEFT JOIN/GROUP BY, and NOT EXISTS queries; no query is executed once per course.

Student Report: GET /api/reports/my

- Returns the authenticated student?s enrolled course count, distinct categories, earliest/latest enrollment dates, total course value, and up to three recent enrollments.
- Both student-report queries use req.user.id as a SQL parameter. The student report response does not include the student?s identity fields.

Dashboard, empty data, and failures

- The admin dashboard requests /api/reports/overview and the student dashboard requests /api/reports/my. The student dashboard continues to load public course availability from the existing statistics endpoint.
- Student dashboard requests are handled independently, so one failed request does not stop successful report data from rendering.
- Zero values and safe date fallbacks prevent empty report data from producing NaN.
- Category bars are built with CSS and guard the zero-maximum case.

Pending enrollments: The current database schema has no enrollment status or pending state. Report totals include all rows in enrollments; pending records cannot be separately identified in this schema.

2.4 Implementation Logic

1. Mount report routes under /api/reports.
2. Authenticate requests and enforce the required role.
3. Run SQL summary and grouped queries for the admin report.
4. Use the JWT-authenticated student ID as a parameter for student report aggregation and recent records.
5. Return normalized report fields and render them in the corresponding dashboard.
6. Render errors and empty states without removing the dashboard structure.

2.5 Error and Data Handling

Controllers log database/report failures and return an explicit HTTP 500 response. Dashboards display request errors, safe zero/date values, and empty states. Student report and course-availability requests are handled independently. Failure injection was not performed in this verification; these paths were reviewed in source.

3. Testing & Verification

3.1 Testing Approach

Verification was performed on 22 October 2026 against the local development application. The active authenticated student session was used for API checks and browser evidence. Frontend lint/build and backend JavaScript syntax checks were also run. Admin-role success and independent SQL comparisons were not run in this verification.

Environment: Local development application (frontend on 127.0.0.1:5173; API on localhost:3000).

Browser: Integrated browser on the local Vite application. Test role: authenticated Student. Administrator success-path verification was unavailable in the active session.

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Request GET /api/reports/my without a token.	Request is rejected as unauthenticated.	HTTP 401 returned.	Pass
TC-02	Request GET /api/reports/overview using the authenticated student session.	Student role is denied administrator report access.	HTTP 403 returned.	Pass
TC-03	Request GET /api/reports/my using the authenticated student session.	Student metrics and recent enrollments are returned.	HTTP 200; 4 enrolled courses, 3 categories, dates present, and 3 recent rows.	Pass
TC-04	Inspect the student report response fields.	Response contains no student identity fields or another student identifier.	Returned fields contained report metrics and course rows only; no student ID/name fields.	Pass
TC-05	Open the Student Dashboard and inspect report values.	Learning summary, dates, and recent enrollments render.	4 courses, Rs. 94,000.00, 3 categories, dates, and 3 recent course rows displayed.	Pass
TC-06	Run frontend lint and production build.	Lint and build complete successfully.	npm run lint and npm run build passed.	Pass
TC-07	Run backend JavaScript syntax and git whitespace checks.	Syntax checks and git diff --check complete successfully.	All node --check commands and git diff --check passed.	Pass

3.3 Test Summary

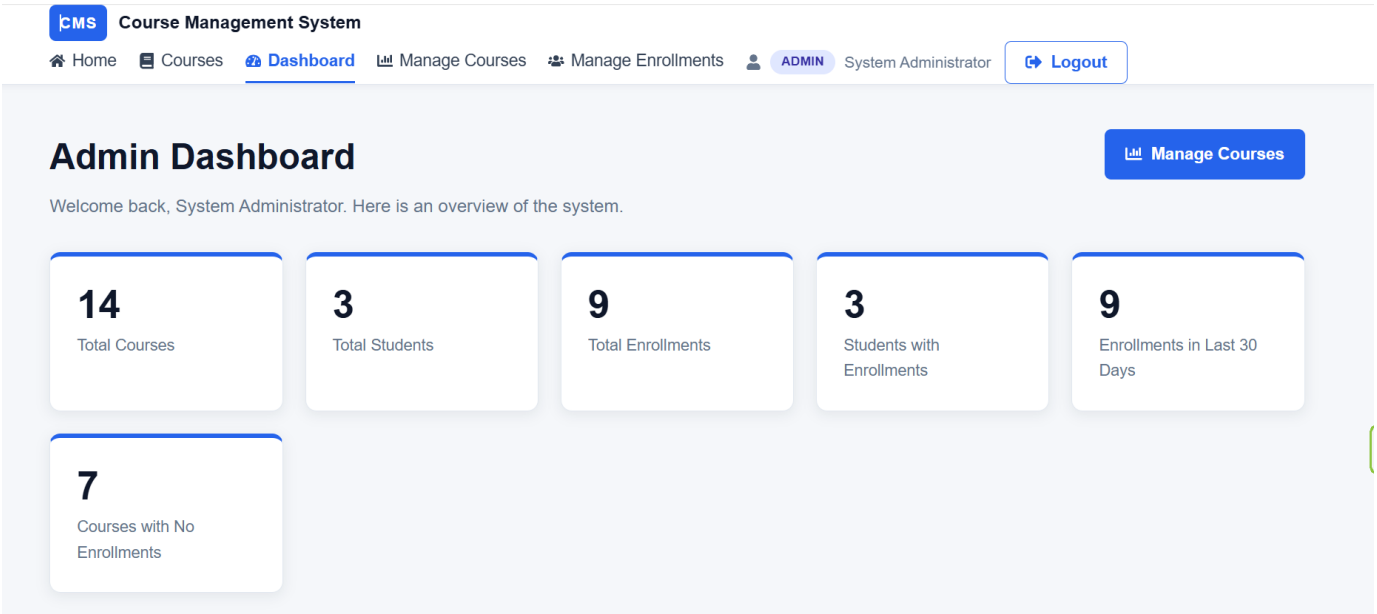
Test Result	**Count**
Total Test Cases	7
Passed	7
Failed	0
Not Run	0
Pass Rate (executed checks)	100% (7/7)

4. Acceptance Criteria Verification

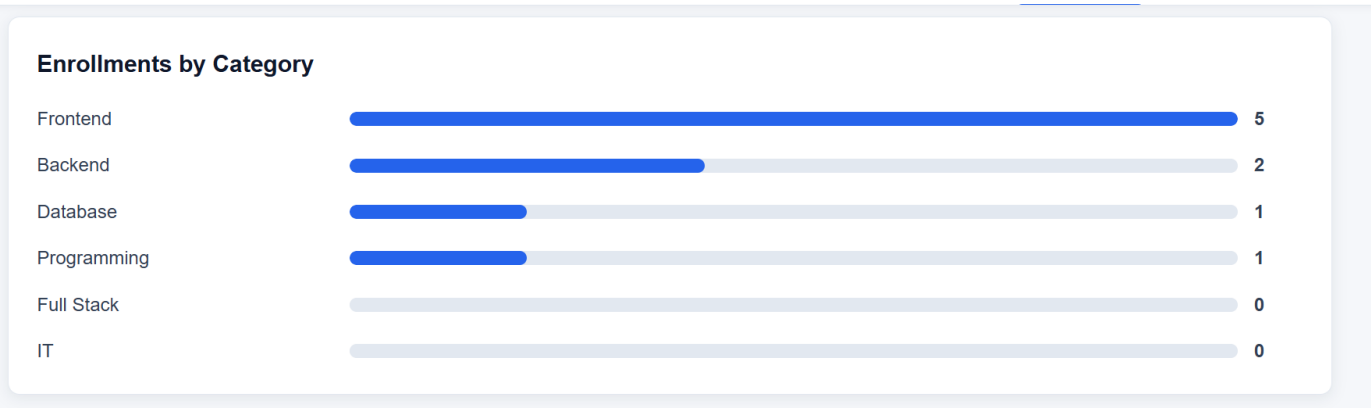
ID	Acceptance Criterion	Verification	Status
AC-001	Admin report returns the required system totals and report sections.	Implementation and route inspected; admin success-path test not run.	Met
AC-002	Only administrators can access the admin report.	Student-role request to admin endpoint returned HTTP 403; route includes admin role guard.	Met
AC-003	Student report returns only the logged-in student?s information.	Authenticated student request returned expected metrics; route scopes queries to req.user.id; response fields inspected.	Met
AC-004	Category totals and top-five course ordering are correct.	SQL grouping, order, and limit inspected; independent SQL comparison not run.	Met
AC-005	Report statistics use SQL aggregation without one query per course.	Report model uses fixed aggregate/grouped query sets; no per-course query loop.	Met
AC-006	Both dashboards use their role-specific reporting endpoints.	Student dashboard verified in browser; admin dashboard integration inspected in source.	Met
AC-007	Top-five ordering and empty values are handled safely.	SQL ordering/limit and zero-safe category/date display logic inspected.	Met
AC-008	Reporting failures do not break the entire dashboard.	Error handling and independent student requests inspected; forced failure not injected.	Met
AC-009	Pending-enrollment handling is defined when supported.	Database schema has no enrollment status or pending state.	Met

5. Evidence

Evidence 01 - Student Dashboard Summary. Related: AC-003, AC-006. The screenshot shows enrolled-course count, total course value, categories explored, course availability, and first/latest enrollment dates.



Evidence 02 - Recent Enrollments, Top Enrollments, Enrollment by Category



Top Courses by Enrollment

COURSE	CATEGORY	ENROLLMENTS
JavaScript	Frontend	2
React Fundamentals	Frontend	2
Advanced Python Programming	Programming	1
Express.js	Backend	1
MySQL	Database	1

Recent Enrollments

Manage enrollments

STUDENT	USERNAME	COURSE	CATEGORY	LEVEL	ENROLLED ON
Nimal Silva	nimal	React Fundamentals	Frontend	BEGINNER	10/6/2026
Kamal Perera	kamal	React Fundamentals	Frontend	BEGINNER	10/6/2026
Kamal Perera	kamal	Advanced Python Programming	Programming	ADVANCED	10/6/2026
Saman Fernando	saman	React	Frontend	ADVANCED	9/22/2026
Saman Fernando	saman	JavaScript	Frontend	INTERMEDIATE	9/22/2026
Nimal Silva	nimal	Express.js	Backend	ADVANCED	9/22/2026

6. Final Result

Item	Result
Requirements	Implemented
Implementation	Completed
Testing	completed
Acceptance Criteria	completed
Dashboard Integration	completed
Final Status	Completed (verification partial)

6.1 Final Summary

CR-009 was implemented by adding role-specific reporting endpoints backed by SQL aggregation and updating both dashboards to consume the appropriate report. The admin report includes system totals, enrollment categories, top-five courses, courses without enrollments, and recent enrollments. The student report is scoped to the authenticated student and returns learning metrics and recent enrollments. All seven checks executed in this verification passed, including student API access, role denial, dashboard rendering, frontend lint/build, backend syntax, and whitespace validation.

Student Name: MSF.Shimla

Student ID: JF/ICT/24/14

Date: 22 October 2026

Signature: _____

~ END OF REPORT ~